

OMAK, WA, USA

WMO: 727890

Lat: 48.464N

Lon: 119.517W

Elev: 396

StdP: 96.66

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 94197

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.2	-11.7	-18.9	0.7	-13.2	-16.4	0.9	-10.3	10.1	-5.9	8.9	-5.7	2.6	0	0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	36.4	17.9	34.5	17.4	32.6	17.0	19.3	32.7	18.4	31.7	17.6	30.4	3.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	11.1	21.3	13.6	10.2	21.0	12.5	9.5	20.9	56.2	32.9	53.3	31.6	50.7	30.4	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
9.0	7.9	7.1	-17.8	39.4	3.3	1.4	-20.1	40.4	-22.0	41.2	-23.8	42.0	-26.2	43.0		
			-18.1	20.5	3.2	1.6	-20.4	21.6	-22.3	22.5	-24.1	23.4	-26.4	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-2.5	-0.2	5.2	9.6	14.5	18.2	23.3	22.8	17.2	9.4	2.3	-2.5	(6)
(7)		DBStd	9.85	4.50	3.83	3.69	3.53	3.87	3.68	3.60	3.40	3.88	3.90	4.39	4.27	(7)
(8)		HDD10.0	1560	388	285	152	49	7	1	0	0	1	56	232	389	(8)
(9)		HDD18.3	3487	646	518	407	261	128	45	5	6	66	278	481	647	(9)
(10)		CDD10.0	1501	0	0	4	38	148	247	412	396	217	38	1	0	(10)
(11)		CDD18.3	387	0	0	0	0	10	42	159	143	32	1	0	0	(11)
(12)		CDH23.3	5651	0	0	0	17	243	696	2264	1927	493	12	0	0	(12)
(13)		CDH26.7	2737	0	0	0	2	71	279	1217	984	182	2	0	0	(13)
(14)	Wind	WSAvg	3.0	2.1	2.6	3.1	3.4	3.3	3.4	3.6	3.5	3.4	2.9	2.5	2.2	(14)
(15)	Precipitation	PrecAvg	290	28	30	23	24	25	22	11	17	14	19	35	47	(15)
(16)		PrecMax	388	64	75	46	62	70	81	36	53	38	61	91	118	(16)
(17)		PrecMin	210	3	5	1	1	1	1	0	0	0	0	0	7	(17)
(18)		PrecStd	49	19	21	14	19	19	19	11	17	15	19	23	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	11.6	19.0	25.9	31.4	36.1	38.9	38.0	34.2	25.1	14.2	8.0	(19)
(20)			MCWB	4.1	5.9	10.2	13.8	16.8	18.2	18.7	18.6	17.0	14.4	9.5	5.0	(20)
(21)		2%	DB	5.2	8.9	16.5	22.5	28.7	32.7	37.1	36.2	31.3	21.1	11.7	5.9	(21)
(22)			MCWB	2.7	4.9	8.4	11.9	15.6	17.1	18.3	17.8	16.2	12.1	7.9	3.8	(22)
(23)		5%	DB	3.1	7.0	14.0	20.2	26.1	29.8	35.2	34.0	28.7	18.4	10.0	3.8	(23)
(24)			MCWB	1.8	3.6	7.3	10.7	14.4	15.9	17.6	17.3	15.3	10.9	7.2	2.2	(24)
(25)		10%	DB	2.2	5.0	12.2	17.8	23.4	27.4	33.0	32.3	26.2	16.5	8.4	2.4	(25)
(26)			MCWB	1.4	2.7	6.4	9.3	13.2	15.2	17.2	16.8	14.2	9.8	6.1	1.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	7.4	11.1	14.4	17.9	19.5	20.8	20.2	18.5	15.1	10.4	5.8	(27)
(28)			MCDWB	7.2	10.1	17.8	24.8	28.8	32.3	34.3	34.6	30.7	22.0	13.0	7.5	(28)
(29)		2%	WB	3.1	5.4	9.2	12.5	16.4	18.0	19.6	19.0	17.0	13.0	8.9	4.0	(29)
(30)			MCDWB	4.5	7.9	15.2	21.1	26.4	30.3	33.1	32.7	29.0	18.2	10.7	5.6	(30)
(31)		5%	WB	2.3	4.0	8.0	11.2	15.2	16.9	18.7	18.0	15.8	11.8	7.8	2.5	(31)
(32)			MCDWB	3.2	6.3	13.1	19.1	24.1	28.0	32.3	31.5	27.0	17.2	9.3	3.5	(32)
(33)		10%	WB	1.4	2.9	6.9	9.9	14.0	15.8	17.8	17.3	14.8	10.5	6.5	1.4	(33)
(34)			MCDWB	2.0	4.7	11.4	16.7	22.2	25.5	31.0	30.2	24.8	15.4	8.2	2.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	8.0	11.5	14.0	14.8	15.0	17.1	17.0	15.9	12.4	7.7	5.2	(35)
(36)			MCDBR	6.4	10.3	14.5	17.3	18.7	18.7	19.7	19.4	19.3	14.9	9.6	7.1	(36)
(37)			MCWBR	5.0	6.9	8.6	8.9	8.4	7.2	6.9	7.1	8.2	8.0	6.5	5.5	(37)
(38)		5% WB	MCDBR	5.7	9.0	13.6	16.2	16.9	17.2	18.2	17.7	17.7	13.3	8.0	6.0	(38)
(39)			MCWBR	4.5	6.3	8.3	8.7	7.9	7.2	6.7	6.6	7.9	7.6	5.8	4.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.282	0.285	0.304	0.326	0.347	0.348	0.341	0.344	0.320	0.314	0.296	0.280	(40)	
(41)		taud	2.354	2.407	2.452	2.450	2.429	2.459	2.490	2.473	2.539	2.538	2.508	2.398	(41)	
(42)		Ebn at Noon	788	881	918	927	914	912	914	897	891	838	778	745	(42)	
(43)		Edh at Noon	68	82	93	103	110	107	103	100	85	72	60	58	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.07	2.11	3.28	4.77	5.84	6.31	7.21	6.04	4.42	2.61	1.25	0.84	(44)	
(45)		RadStd	0.13	0.23	0.27	0.35	0.43	0.49	0.42	0.39	0.32	0.29	0.16	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83	+34

Nomenclature: See separate page